

TOUR DE MATHS TRAINING

1. m and n are integers with $m > n$. The number of integers between (but not including m and n) is

- (a) $m - n$ (b) $m - n - 1$ (c) $m - n + 1$ (d) $m + n - 1$

Solution (b): If m and n are included, the number is $m - (n - 1) = m - n + 1$. But they are not, so the answer is $m - n + 1 - 2 = m - n - 1$.

2. Successive discounts of 12% and 25% is the same as a single discount of

- (a) 34% (b) 35% (c) 36% (d) 37%

Solution (a)

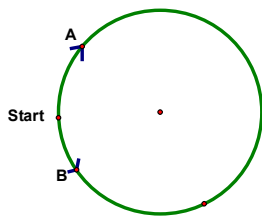
We can assume the article purchased is marked at R100. A discount of 12% brings it to R88 and a further discount of 25% reduces it by R22, to $R88 - R22 = R66$. So the combined discount is R34, that is 34%.

3. A and B run a race of $\frac{4}{5}$ km on a circular track having circumference $\frac{1}{4}$ km. They start together

but run in opposite directions. If A wins by 40 metres, how far was B from the starting post when A passed it the first time?

- (a) 12,5 metres (b) 15 metres (c) 17,5 metres (d) 20 metres

Solution: (a)



They have to cover 800m and the track has length 250m. When A completes 800m, B completes 760 metres. Suppose A runs in a clockwise direction. So when A completes 250 metres, B runs

$\frac{760}{800} \times 250 = \frac{76 \times 25}{8} = \frac{19 \times 25}{2} = \frac{475}{2} = 237,5$ metres. So he is $250 - 137,5 = 112,5$ metres from the start.

4. The highest vehicle registration number (VRN) in Durban is just under ND 840000. The *sum total* of a VRN is the sum of the digits in the number, Thus the sum total of ND 678645 is

$$6 + 7 + 8 + 6 + 4 + 5 = 36.$$

How many VRNs in Durban have sum total greater than 49?

- (a) 13 (b) 24 (c) 26 (d) 28

Solution: (d)

The highest possible sum is from 799999, which has sum 52. There is only one vehicle having sum 52. Which vehicles have sum 51? To obtain them we reduce each of the six digits of 799999 by 1; there are six of them, namely, 699999, 789999, 798999, 799899, 799989, 799998.

Finally, which vehicles have sum 50? We can reduce each of the 6 digits by 2. This gives 6 VRNs. Or, since $1 + 1 = 2$, we can reduce each *pair of two digits* from 799999 by 1 and leave the rest as is. For example, 689999 is one of them. This VRN corresponds to "12", that is, the first and second digits are each reduced by 1. We can now count as follows:

12 13 14 15 16 23 24 25 26 34 35 36 45 46 56.

For example 56 corresponds to the VRN 799988. There are 15 such pairs. Hence there are

$$1 + 6 + 6 + 15 = 28 \text{ such VRNs.}$$

5. In $\triangle ABC$, $AB = AC$, $BD = BC$, and $\angle A = 2x$.

Then $\angle ABD$ is equal to

- (a) $2x$ (b) $90^\circ - 3x$ (c) $90^\circ - 2x$ (d) $90^\circ - x$

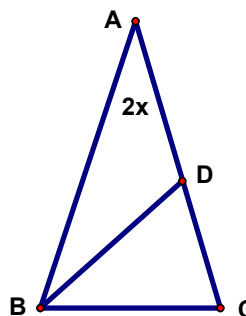
Solution: (b)

Since $AB = AC$, $\angle C = \angle B = \frac{180^\circ - 2x}{2} = 90^\circ - x$.

Since $BD = BC$, $\angle BDC = \angle C = 90^\circ - x$

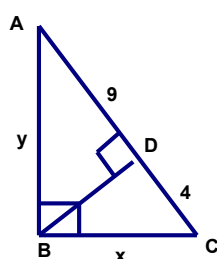
Hence $\angle DBC = 180^\circ - 2(90^\circ - x) = 2x$ (angles of a triangle)

Finally $\angle ABD = (90^\circ - x) - 2x = 90^\circ - 3x$



6. $\angle ABC = \angle ADB = 90^\circ$. What is the length of BD ?

Solution 1:



Let x and y be as shown. Since ABC is a right angle,

$$x^2 + y^2 = (9 + 4)^2 = 169 \dots\dots\dots(1)$$

from Pythagoras. Again from Pythagoras, we can find BD in two different ways, which allows us to create the equation:

$$y^2 - 81 = x^2 - 16$$

That is,

$$x^2 - y^2 = -65 \dots\dots\dots(2)$$

Adding (1) and (2), we obtain $x^2 = 52$ and hence, $BD^2 = x^2 - 16 = 52 - 16 = 36$. That is, $BD = 6$.

Solution 2

$$\angle ABD = 90^\circ - \angle CBD = \angle C$$

$$\tan \angle ABD = \frac{9}{BD} = \tan \angle C = \frac{BD}{4} \text{ (see diagram: tan is opposite over adjacent)}$$

Hence $BD^2 = 36$ and $BD = 6$.

7. The figure shows two concentric circles. If the area of the ring between the two circles is 144π square centimetres, what is the length of PQ , in centimetres?

Solution: 24

$144\pi =$ the area of the large circle – area of the small circle

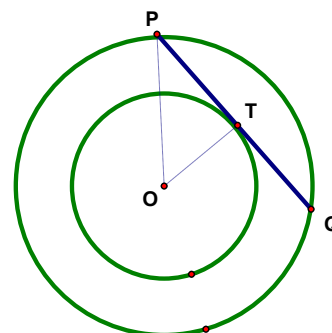
$$= \pi OP^2 - \pi OT^2 = \pi(OP^2 - OT^2)$$

$$\text{So } 144 = OP^2 - OT^2$$

Now, the radius of a circle is always perpendicular to a tangent at the point of contact. So OT is perpendicular to PQ . So from Pythagoras, $OP^2 - OT^2 = PT^2$. that is, $144 = PT^2$, and $PT = 12$.

But we also know that since OT is perpendicular to PQ , $PT = TQ$.

Hence $PQ = 24$.



8. When $333\,333\,333\,333\,333\,333^2 + 222\,222\,222\,222\,222\,222$ is written out in full what is the sum of its digits?

Solution: 36

First note that each of the numbers contains 18 digits and the second is $\frac{2}{3}$ of the first. Let

$x = 333\,333\,333\,333\,333\,333$. Then our number is $x^2 + \frac{2}{3}x = \frac{3x^2 + 2x}{3} = \frac{x(3x+2)}{3}$. Now $3x+2$ is a number that 2 more than one that has 18 nines. That is, it is 1 followed by 17 zeros and then by a 1, which is $10^{18} + 1$. So

$$x(3x+2) = x(10^{18}) + x$$

This number is x followed by 18 zeros – it has 36 digits - added to x , which has 18 digits. Since x has only 3's, $x(3x+2)$ is a 36 digit number consisting only of 3's. Finally our number $\frac{x(3x+2)}{3}$ is a 36-digit number consisting only of 1's. The sum of the digits is 36.

9. Four teams A, B, C and D each play a single game against each of the other teams. There are no draws and teams A, B C have the same number of wins. If team A beats team D, how many wins does team D have?

Solution: None

Six games were played altogether, namely, AB, AC, AD, BC, BD, CD. So there were six wins.

Suppose A, B and C each won x games, and that D won y games. The $3x + y = 6$. Now

$0 \leq y < 3$ since D lost a game. Substitute $y = 0, 1$ and 2 . The solutions for x are $2, \frac{5}{3}, \frac{4}{3}$. Since

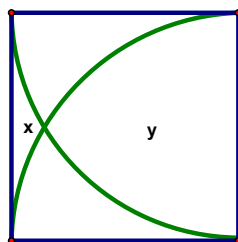
x is an integer, $x = 2$ and $y = 0$. So D won no games.

Alternative solution: Draw a table. Give 2 for a win, 1 for a draw and 0 for a loss. We have

	A	B	C	D
A	X			2
B		X		
C			X	
D	0			X

Six games were played altogether. So there are 6 twos, 0 ones and 6 zeros in the table, giving a total sum of 12. The rows A, B and C each have sum 2 (if they win exactly 1 game each) or sum 4 (if they win 2 games each). In the first case, D will have sum 6 which is impossible. Hence each of A, B and C has sum 4, forcing D to have sum 0. That is, D won no games.

10. **Solution** : $2\pi - 4$



The diagram shows a square and two quarter circles. Each quarter circle is centred on the vertex of the square. If a side of the square has length 2, and x and y are the areas of the indicated regions, what is $y - x$ equal to?

Solution: By symmetry, the remaining two areas are equal. Let each of these have area z . Then

$$x + y + 2z = \text{area of square} = 4$$

$$y + z = \text{area of quarter circle} = \frac{1}{4}\pi(2^2) = \pi$$

So $z = \pi - 4$. Substitute in the first equation:

$$x + y + 2(\pi - y) = 4$$

$$x + y + 2\pi - 2y = 4$$

$$y - x = 2\pi - 4$$